

SECTION 10 ELECTRICAL CHARACTERISTICS

This section provides the thermal characteristics and electrical specifications for the MC68020/EC020. Note that the thermal and DC electrical characteristics are listed separately for the MC68020 and the MC68EC020. All other data applies to both the MC68020 and the MC68EC020 unless otherwise noted.

10.1 MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	−0.3 to +7.0	V
Input Voltage	V_{in}	−0.5 to +7.0	V
Operating Temperature Range			
Minimum Ambient Temperature	T_A	0	°C
Maximum Ambient Temperature PGA, PPGA, PQFP	T_A	70	°C
Maximum Junction Temperature CQFP	T_J	110	°C
Storage Temperature Range	T_{stg}	−55 to 150	°C

The device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, normal precautions should be taken to avoid application of voltages higher than maximum-rated voltages to these high-impedance circuits. Tying unused inputs to the appropriate logic voltage level (e.g., either GND or V_{CC}) enhances reliability of operation.

10.2 THERMAL CONSIDERATIONS

The average chip-junction temperature, T_J , in °C can be obtained from:

$$T_J = T_A + (P_D \cdot \theta_{JA}) \quad (10-1)$$

where:

- T_A = Ambient Temperature, °C
- θ_{JA} = Package Thermal Resistance, Junction-to-Ambient, °C/W
- P_D = $P_{INT} + P_{I/O}$
- P_{INT} = $I_{CC} \times V_{CC}$, Watts—Chip Internal Power
- $P_{I/O}$ = Power Dissipation on Input and Output Pins—User Determined

For most applications, $P_{I/O} < P_{INT}$ and can be neglected.

An approximate relationship between P_D and T_J (if $P_{I/O}$ is neglected) is:

$$P_D = K \div (T_J + 273^\circ\text{C}) \quad (10-2)$$

Solving Equations (10-1) and (10-2) for K gives:

$$K = P_D \cdot (T_A + 273^{\circ}\text{C}) + \theta_{JA} \cdot P_D^2 \quad (10-3)$$

where K is a constant pertaining to the particular part. K can be determined from equation (10-3) by measuring P_D (at thermal equilibrium) for a known T_A . Using this value of K, the values of P_D and T_J can be obtained by solving equations (10-1) and (10-2) iteratively for any value of T_A .

The total thermal resistance of a package (θ_{JA}) can be separated into two components, θ_{JC} and θ_{CA} . θ_{JC} represents the barrier to heat flow from the semiconductor junction to the package (case) surface, and θ_{CA} represents the barrier to heat flow from the case to the ambient air. These terms are related by the equation:

$$\theta_{JA} = \theta_{JC} + \theta_{CA} \quad (10-4)$$

θ_{JC} is device related and cannot be influenced by the user. However, θ_{CA} is user dependent and can be minimized by such thermal management techniques as heat sinks, forced air cooling, and use of thermal convection to increase air flow over the device. Thus, good thermal design on the part of the user can significantly reduce θ_{CA} so that θ_{JA} approximately equals θ_{JC} . Substitution of θ_{JC} for θ_{JA} in equation (10-1) results in a lower semiconductor junction temperature.

10.2.1 MC68020 Thermal Characteristics and DC Electrical Characteristics

MC68020 Thermal Resistance ($^{\circ}\text{C}/\text{W}$)

The following table provides thermal resistance characteristics for junction to ambient and junction to case for the MC68020 packages with natural convection and no heatsink.

Characteristic— Natural Convection and No Heatsink	θ_{JA}	θ_{JC}
Thermal Resistance		
PGA Package (RC Suffix)	26	3
PPGA Package (RP Suffix)	32	10
CQFP Package (FE Suffix)	46	15
PQFP Package (FC Suffix)	42	20

Resistance is to bottom center (pin side) of case for PGA and PPGA packages, top center of case for CQFP and PQFP packages.

MC68020 CQFP Package

Table 10-1 provides typical and worst case thermal characteristics for the MC68020 CQFP package both with and without a heatsink. The heatsink used is black anodized aluminum alloy, 0.72"x0.75"x0.6" high with an omnidirectional 5x6 array of fins. Attachment was made using Epolite 6400 one part epoxy.

Table 10-1. θ_{JA} vs. Airflow—MC68020 CQFP Package

θ_{JA}	Airflow in Linear Feet/Minute		
	0*	200	500
Maximum			
No Heatsink	46	28	24
With Heatsink	35	20	18
Typical			
No Heatsink	43	25	21
With Heatsink	32	17	15

*Natural convection

Table 10-2 shows the relationship between clock speed and power dissipation for any package type. The worst case operating conditions are used for thermal management design, while typical values are used for reliability analysis.

**Table 10-2. Power vs. Rated Frequency
(at T_J Maximum = 110°C)**

Rated Frequency (MHz)	P_D Maximum (Watts)	P_D Typical (Watts)
33	1.4	0.84
25	1.2	0.72
20	1.0	0.60
16	0.9	0.54

Table 10-3 shows the relationship between board temperature rise and power dissipation in the test environment for the CQFP package. Derate θ_{JA} based on measurements made in the application by adding $(0.8/P_D) * [T_{ba(application)} - T_{ba(table)}]$ to the θ_{JA} values in the table. Board temperature was measured on the top surface of the board directly under the device.

**Table 10-3. Temperature Rise of Board vs. P_D
—MC68020 CQFP Package**

Natural Convection	P_D		
	0.6W	1.0W	1.75W
T_{ba} (°C)—No Heatsink	18	27	53

Values for thermal resistance presented in this document were derived using the procedure described in Motorola Reliability Report 7843, "Thermal Resistance Measurement Method for MC68XX Microcomponent Devices," and are provided for design purposes only. Thermal measurements are complex and dependent on procedure and setup. User-derived values for thermal resistance may differ.

MC68020 DC Electrical Characteristics

($V_{CC} = 5.0 V_{dc} \pm 5\%$; $GND = 0 V_{dc}$; Temperature within defined ranges)

Characteristics	Symbol	Min	Max	Unit
Input High Voltage	V_{IH}	2.0	V_{CC}	V
Input Low Voltage	V_{IL}	GND -0.5	0.8	V
Input Leakage Current BERR, BR, BGACK, CLK, IPL2-IPL0, AVEC, DSACK1, DSACK0, CDIS GND $\leq V_{in} \leq V_{CC}$ HALT, RESET	I_{in}	-1.0 -20	1.0 20	μA
Hi-Z (Off-State) Leakage Current A31-A0, AS, DBEN, DS, D31-D0, FC2-FC0, R/W, RMC, SIZ1-SIZ0 @ 2.4 V/0.5 V	I_{TSI}	-20	20	μA
Output High Voltage A31-A0, AS, BG, D31-D0, DBEN, DS, R/W, ECS, IPEND, RMC, SIZ1-SIZ0, FC2-FC0 $I_{OH} = 400 \mu A$	V_{OH}	2.4	—	V
Output Low Voltage A31-A0, FC2-FC0, SIZ1-SIZ0, BG, D31-D0 $I_{OL} = 3.2 \text{ mA}$ $I_{OL} = 5.3 \text{ mA}$ $I_{OL} = 2.0 \text{ mA}$ $I_{OL} = 10.7 \text{ mA}$ AS, DS, R/W, RMC, DBEN, IPEND, ECS, OCS HALT, RESET	V_{OL}	— — — —	0.5 0.5 0.5 0.5	V
Power Dissipation ($T_A = 0^\circ C$)	P_D	—	2.0	W
Capacitance (see Note) $V_{in} = 0 V$, $T_A = 25^\circ C$, $f = 1$ MHz	C_{in}	—	20	pF
Load Capacitance ECS, OCS All Other	C_L	— —	50 130	pF

NOTE: Capacitance is periodically sampled rather than 100% tested.

10.2.2 MC68EC020 Thermal Characteristics and DC Electrical Characteristics

MC68EC020 Thermal Resistance ($^\circ C/W$)

The following table provides thermal resistance characteristics for junction to ambient and junction to case for the MC68EC020 packages with natural convection and no heatsink.

Characteristic – Natural Convection and No Heatsink	θ_{JA}	θ_{JC}
Thermal Resistance		
PPGA Package (RP Suffix)	32	10
PQFP Package (FG Suffix)	53	18

MC68EC020 PQFP Package

Table 10-4 provides typical and worst case thermal characteristics for the MC68EC020 PQFP package without a heatsink.

Table 10-4. θ_{JA} vs. Airflow – MC68EC020 PQFP Package

θ_{JA}	Airflow in Linear Feet/Minute					
	0*	50	100	200	300	400
Maximum—No Heatsink	53	49	45	41	38	36
Typical—No Heatsink	51	47	43	39	37	35

MC68EC020 DC Electrical Characteristics(V_{CC} = 5.0 V_{dc} ± 5%; GND = 0 V_{dc}; Temperature within defined ranges)

Characteristics	Symbol	Min	Max	Unit
Input High Voltage	V _{IH}	2.0	V _{CC}	V
Input Low Voltage	V _{IL}	GND -0.5	0.8	V
Input Leakage Current GND ≤ V _{in} ≤ V _{CC} BERR, BR, CLK, IPL2-IPL0, AVEC, DSACK1, DSACK0, CDIS, HALT, RESET	I _{in}	-1.0 -20	1.0 20	μA
Hi-Z (Off-State) Leakage Current @ 2.4 V/0.5 V A23-A0, AS, DS, D31-D0, FC2-FC0, R/W, RMC, SIZ1-SIZ0	I _{TSI}	-20	20	μA
Output High Voltage I _{OH} = 400 μA A23-A0, AS, BG, D31-D0, DS, R/W, RMC, SIZ1-SIZ0, FC2-FC0	V _{OH}	2.4	—	V
Output Low Voltage I _{OL} = 3.2 mA I _{OL} = 5.3 mA I _{OL} = 10.7 mA A23-A0, FC2-FC0, SIZ1-SIZ0, BG, D31-D0, AS, DS, R/W, RMC, HALT, RESET	V _{OL}	— — —	0.5 0.5 0.5	V
Power Dissipation (T _A = 0°C) f = 25 MHz f = 16 MHz	P _{INT}	— —	1.5 1.2	W
Capacitance (see Note) V _{in} = 0 V, T _A = 25°C, f = 1 MHz	C _{in}	—	20	pF
Load Capacitance	C _L	—	130	pF

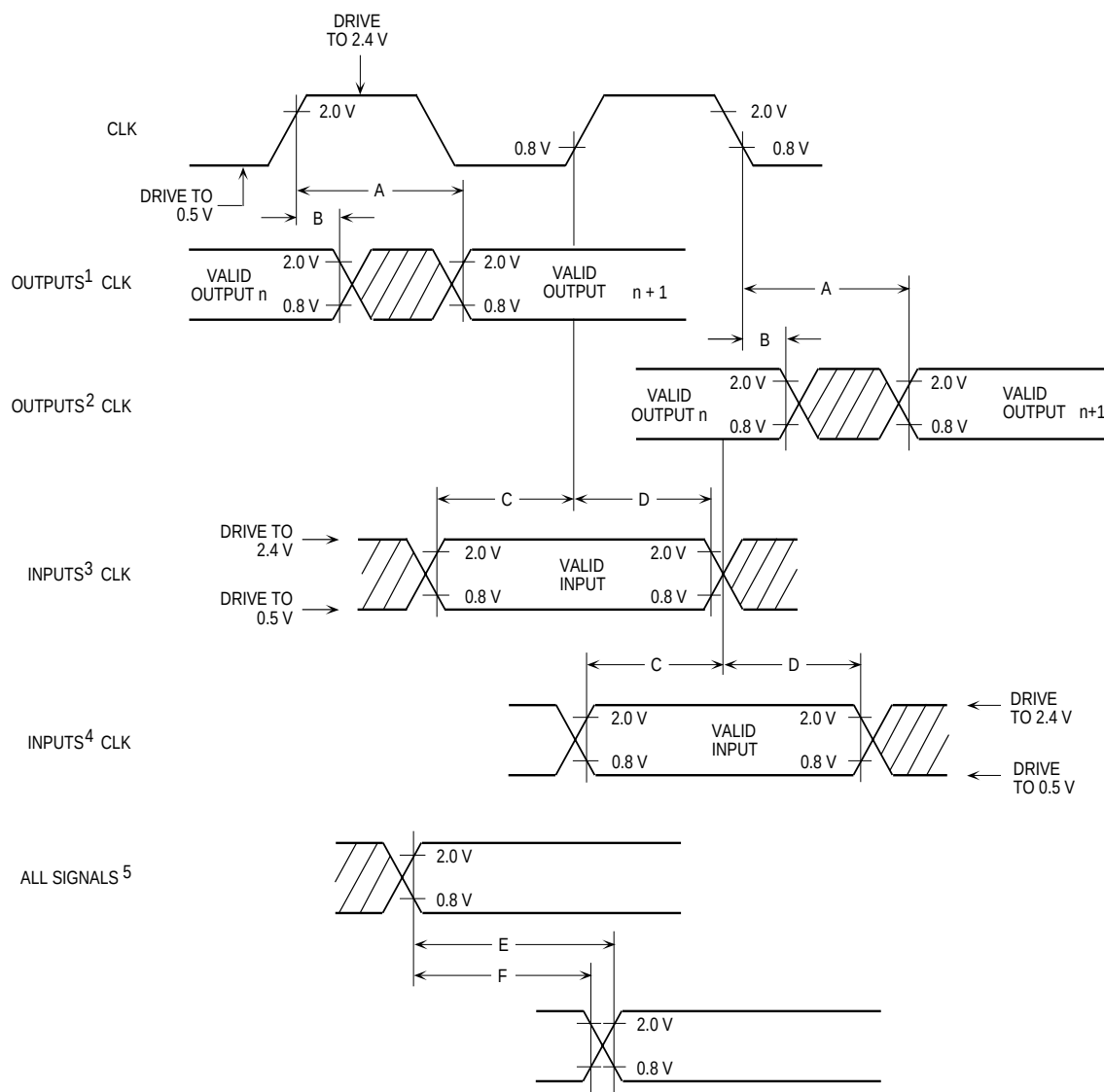
NOTE: Capacitance is periodically sampled rather than 100% tested.

10.3 AC ELECTRICAL CHARACTERISTICS

The AC specifications presented consist of output delays, input setup and hold times, and signal skew times. All signals are specified relative to an appropriate edge of the clock and possibly to one or more other signals.

The measurement of the AC specifications is defined by the waveforms shown in Figure 10-1. To test the parameters guaranteed by Motorola, inputs must be driven to the voltage levels specified in Figure 10-1. Outputs are specified with minimum and/or maximum limits, as appropriate, and are measured as shown in Figure 10-1. Inputs are specified with minimum setup and hold times, and are measured as shown. Finally, the measurement for signal-to-signal specifications is also shown.

Note that the testing levels used to verify conformance to the AC specifications do not affect the guaranteed DC operation of the device as specified in the DC electrical specifications. The 20 MHz and 33.33 MHz specifications do not apply to the MC68EC020.



NOTES:

1. This output timing is applicable to all parameters specified relative to the rising edge of the clock. □
2. This output timing is applicable to all parameters specified relative to the falling edge of the clock. □
3. This input timing is applicable to all parameters specified relative to the rising edge of the clock. □
4. This input timing is applicable to all parameters specified relative to the falling edge of the clock. □
5. This timing is applicable to all parameters specified relative to the assertion/negation of another signal.

LEGEND:

- A. Maximum output delay specification. □
- B. Minimum output hold time. □
- C. Minimum input setup time specification. □
- D. Minimum input hold time specification. □
- E. Signal valid to signal valid specification (maximum or minimum). □
- F. Signal valid to signal invalid specification (maximum or minimum).

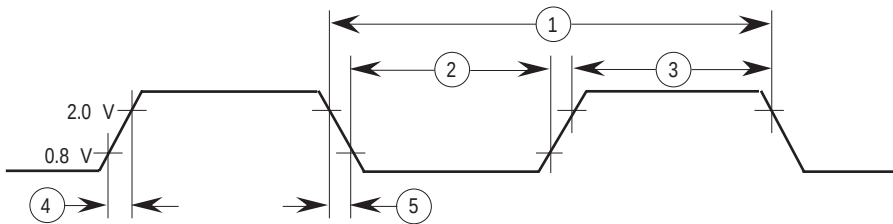
FIGURE 10-1
MC68020UM

Figure 10-1. Drive Levels and Test Points for AC Specifications

AC ELECTRICAL CHARACTERISTICS—CLOCK INPUT (see Figure 10-2)

Num.	Characteristic	16.67 MHz		20 MHz		25 MHz*		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
	Frequency of Operation	8	16.67	12.5	20	12.5	25	12.5	33.33	MHz
1	Cycle Time	60	125	50	80	40	80	30	80	ns
2,3	Clock Pulse Width (Measured from 1.5 V to 1.5 V)	24	95	20	54	19	61	14	66	ns
4,5	Clock Rise and Fall Times	—	5	—	5	—	4	—	3	ns

*These specifications represent an improvement over previously published specifications for the 25-MHz MC68020 and are valid only for products bearing date codes of 8827 and later.



NOTE: Timing measurements are referenced to and from a low voltage of .08 V and a high voltage of 2.0 V, unless otherwise noted. The voltage swing through this range should start outside and pass through the range such that the rise or fall is between 0.8 V and 2.0 V.

FIGURE 10-2
MC68020UM

Figure 10-2. Clock Input Timing Diagram

AC ELECTRICAL CHARACTERISTICS—READ AND WRITE CYCLES(V_{CC} = 5.0 V_{dc} ± 5%; GND = 0 V_{dc}; Temperature within defined ranges; see Figures 10-3–10-5)

Num.	Characteristics	16.67 MHz		20 MHz		25 MHz**		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
6	Clock High to FC, Size, RMC, Address Valid	0	30	0	25	0	25	0	21	ns
*6A	Clock High to ECS, OCS Asserted	0	20	0	15	0	12	0	10	ns
7	Clock High to Address, Data, FC, Size, RMC High Impedance	0	60	0	50	0	40	0	30	ns
8	Clock High to Address, FC, Size, RMC Invalid	0	—	0	—	0	—	0	—	ns
9	Clock Low to AS, DS Asserted	3	30	3	25	3	18	3	15	ns
9A ¹	AS to DS Assertion Skew (Read)	–15	15	–10	10	–10	10	–10	10	ns
9B ¹¹	AS Asserted to DS Asserted (Write)	37	—	32	—	27	—	22	—	ns
*10	ECS Width Asserted	20	—	15	—	15	—	10	—	ns
*10A	OCS Width Asserted	20	—	15	—	15	—	10	—	ns
*10B ⁷	ECS, OCS Width Negated	15	—	10	—	5	—	5	—	ns
11	Address, FC, Size, RMC Valid to AS (and DS Asserted, Read)	15	—	10	—	6	—	5	—	ns
12	Clock Low to AS, DS Negated	0	30	0	25	0	15	0	15	ns
*12A	Clock Low to ECS, OCS Negated	0	30	0	25	0	15	0	15	ns
13	AS, DS Negated to Address, FC, Size, RMC Invalid	15	—	10	—	10	—	5	—	ns
14	AS (and DS Read) Width Asserted	100	—	85	—	70	—	50	—	ns
14A	DS Width Asserted (Write)	40	—	38	—	30	—	25	—	ns
15	AS, DS Width Negated	40	—	38	—	30	—	23	—	ns
15A ⁸	DS Negated to AS Asserted	35	—	30	—	25	—	18	—	ns
16	Clock High to AS, DS, R/W, DBEN High Impedance	—	60	—	50	—	40	—	30	ns
17	AS, DS Negated to R/W Invalid	15	—	10	—	10	—	5	—	ns
18	Clock High to R/W High	0	30	0	25	0	20	0	15	ns
20	Clock High to R/W Low	0	30	0	25	0	20	0	15	ns
21	R/W High to AS Asserted (Read)	15	—	10	—	5	—	5	—	ns
22	R/W Low to DS Asserted (Write)	75	—	60	—	50	—	35	—	ns
23	Clock High to Data-Out Valid	—	30	—	25	—	25	—	18	ns
25	AS, DS Negated to Data-Out Invalid	15	—	10	—	5	—	5	—	ns
*25A ⁹	DS Negated to DBEN Negated (Write)	15	—	10	—	5	—	5	—	ns
26	Data-Out Valid to DS Asserted (Write)	15	—	10	—	5	—	5	—	ns
27	Data-In Valid to Clock Low (Setup) (Read)	5	—	5	—	5	—	5	—	ns
27A	Late BERR/HALT Asserted to Clock Low (Setup)	20	—	15	—	10	—	5	—	ns
28	AS, DS Negated to DSACK _≈ , BERR, HALT, AVEC Negated	0	80	0	65	0	50	0	40	ns

AC ELECTRICAL CHARACTERISTICS—READ AND WRITE CYCLES

(Continued)

Num.	Characteristics	16.67 MHz		20 MHz		25 MHz**		33.33 MHz		Unit
		Min	Max	Min	Max	Min	Max	Min	Max	
29	AS, DS Negated to Data-In Invalid (Data-In Hold Time)	0	—	0	—	0	—	0	—	ns
29A	AS, DS Negated to Data-In (High Impedance)	—	60	—	50	—	40	—	30	ns
30	Clock Low to Data-In Invalid (Data-In Hold Time)	15	—	15	—	10	—	10	—	ns
31 ²	DSACK _≈ Asserted to Data-In Valid	—	50	—	43	—	32	—	17	ns
31A ³	DSACK _≈ Asserted to DSACK _≈ Valid (DSACK _≈ Asserted Skew)	—	15	—	10	—	10	—	10	ns
32	RESET Input Transition Time	—	1.5	—	1.5	—	1.5	—	1.5	Clks
33	Clock Low to BG Asserted	0	30	0	25	0	20	0	20	ns
34	Clock Low to BG Negated	0	30	0	25	0	20	0	20	ns
35	BR Asserted to BG Asserted (RMC Not Asserted)	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	Clks
*37	BGACK Asserted to BG Negated	1.5	3.5	1.5	3.5	1.5	3.5	1.5	3.5	Clks
*37A ⁶	BGACK Asserted to BR Negated	0	1.5	0	1.5	0	1.5	0	1.5	Clks
39	BG Width Negated	90	—	75	—	60	—	50	—	ns
39A	BG Width Asserted	90	—	75	—	60	—	50	—	ns
*40	Clock High to DBEN Asserted (Read)	0	30	0	25	0	20	0	15	ns
*41	Clock Low to DBEN Negated (Read)	0	30	0	25	0	20	0	15	ns
*42	Clock Low to DBEN Asserted (Write)	0	30	0	25	0	20	0	15	ns
*43	Clock High to DBEN Negated (Write)	0	30	0	25	0	20	0	15	ns
*44	R/W Low to DBEN Asserted (Write)	15	—	10	—	10	—	5	—	ns
*45 ⁵	DBEN Width Asserted	60	—	50	—	40	—	30	—	ns
	Read	120	—	100	—	80	—	60	—	
46	R/W Width Valid (Write or Read)	150	—	125	—	100	—	75	—	ns
47A	Asynchronous Input Setup Time	5	—	5	—	5	—	5	—	ns
47B	Asynchronous Input Hold Time	15	—	15	—	10	—	10	—	ns
48 ⁴	DSACK _≈ Asserted to BERR, HALT Asserted	—	30	—	20	—	18	—	15	ns
53	Data-Out Hold from Clock High	0	—	0	—	0	—	0	—	ns
55	R/W Valid to Data Bus Impedance Change	30	—	25	—	20	—	20	—	ns
56	RESET Pulse Width (Reset Instruction)	512	—	512	—	512	—	512	—	Clks
57	BERR Negated to HALT Negated (Rerun)	0	—	0	—	0	—	0	—	ns
*58 ¹⁰	BGACK Negated to Bus Driven	1	—	1	—	1	—	1	—	Clks
59 ¹⁰	BG Negated to Bus Driven	1	—	1	—	1	—	1	—	Clks

*This specification does not apply to the MC68EC020.

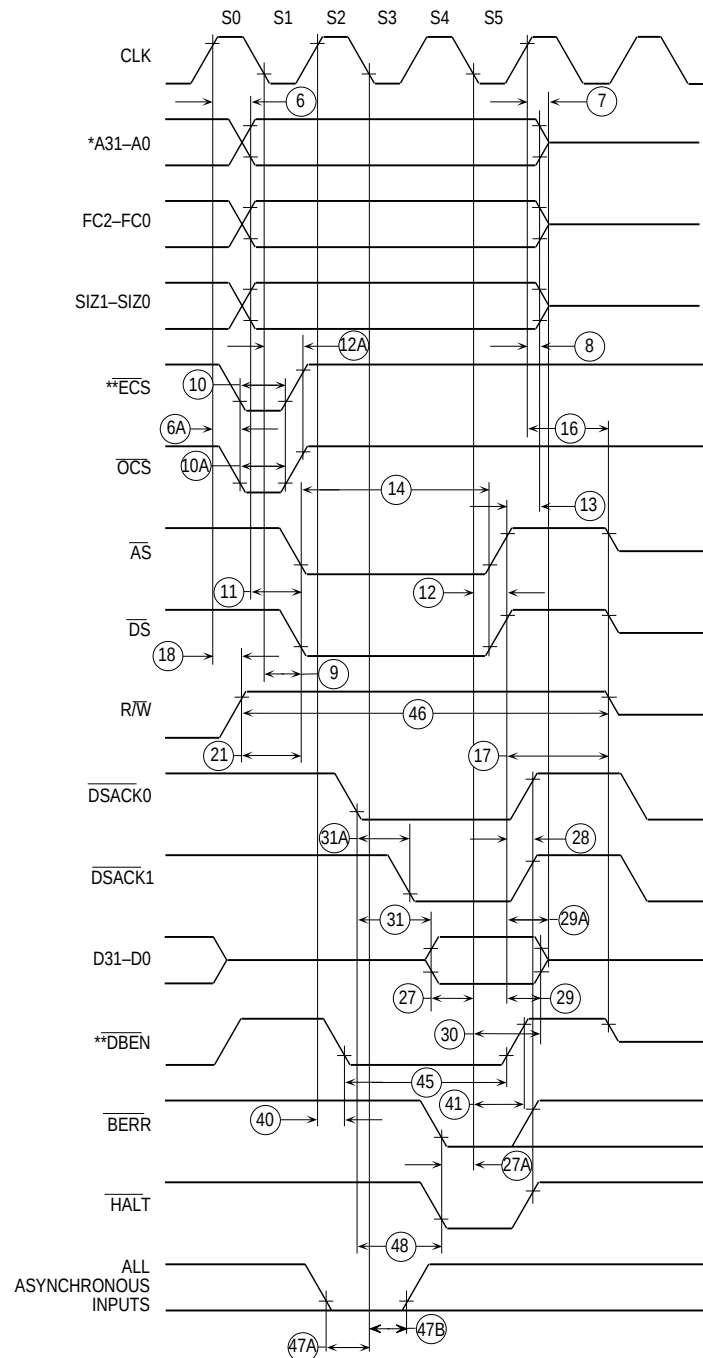
**These specifications represent an improvement over previously published specifications for the 25-MHz MC68020 and are valid only for product bearing date codes of 8827 and later.

AC ELECTRICAL CHARACTERISTICS—READ AND WRITE CYCLES

(Concluded)

NOTES:

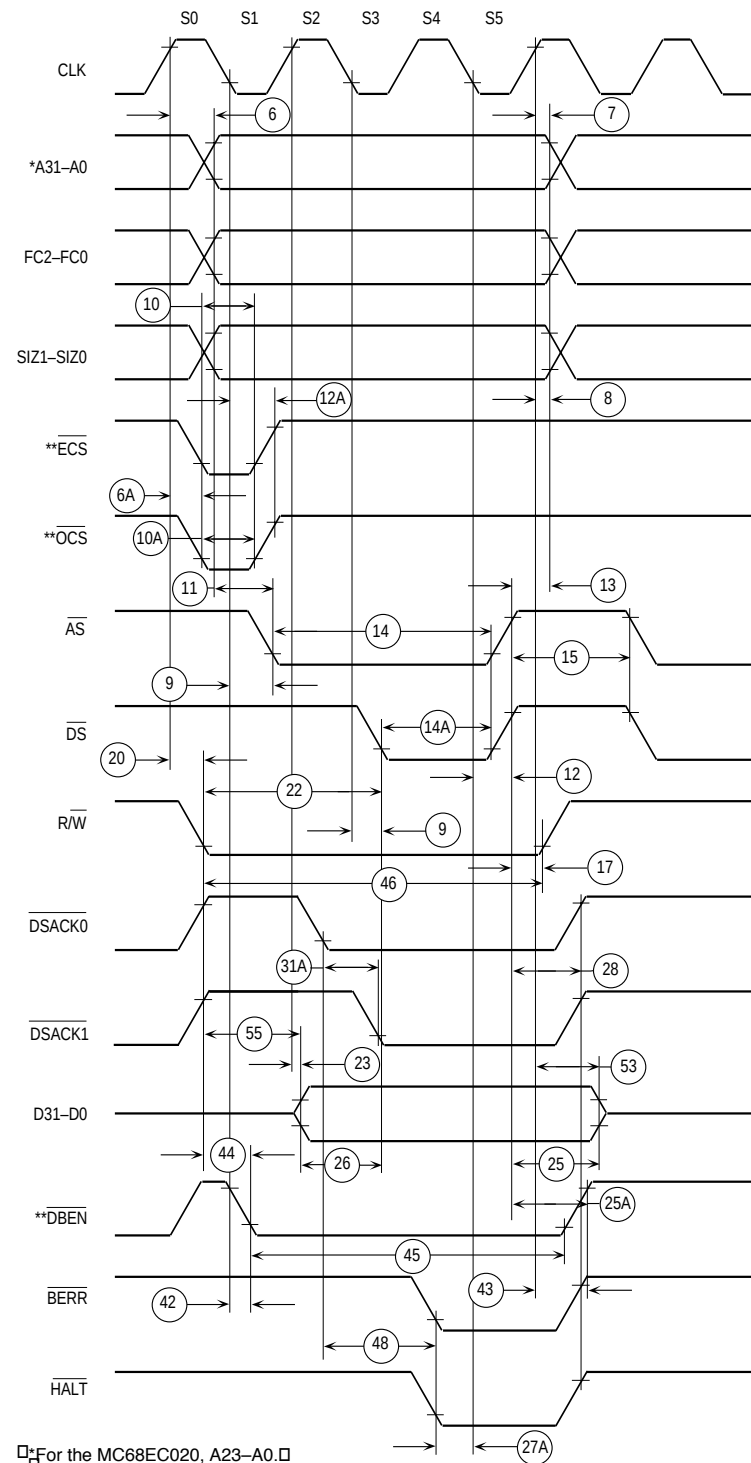
1. This number can be reduced to 5 ns if strobes have equal loads.
2. If the asynchronous setup time (#47A) requirements are satisfied, the DSACK_≈ low to data setup time (#31) and DSACK_≈ low to BERR low setup time (#48) can be ignored. The data must only satisfy the data-in clock low setup time (#27) for the following clock cycle, and BERR must only satisfy the late BERR low to clock low setup time (#27A) for the following clock cycle.
3. This parameter specifies the maximum allowable skew between DSACK0 to DSACK1 asserted or DSACK1 to DSACK0 asserted; specification #47A must be met by DSACK0 or DSACK1.
4. This specification applies to the first (DSACK0 or DSACK1) DSACK_≈ signal asserted. In the absence of DSACK_≈, BERR is an asynchronous input using the asynchronous input setup time (#47A).
5. DBEN may stay asserted on consecutive write cycles.
6. The minimum values must be met to guarantee proper operation. If this maximum value is exceeded, BG may be reasserted.
7. This specification indicates the minimum high time for ECS and OCS in the event of an internal cache hit followed immediately by a cache miss or operand cycle.
8. This specification guarantees operation with the MC68881/MC68882, which specifies a minimum time for DS negated to AS asserted (specification #13A in MC68881UM/AD, MC68881/MC68882 Floating-Point Coprocessor User's Manual). Without this specification, incorrect interpretation of specifications #9A and #15 would indicate that the MC68020/EC020 does not meet the MC68881/MC68882 requirements.
9. This specification allows a system designer to guarantee data hold times on the output side of data buffers that have output enable signals generated with DBEN.
10. These specifications allow system designers to guarantee that an alternate bus master has stopped driving the bus when the MC68020/EC020 regains control of the bus after an arbitration sequence.
11. This specification allows system designers to qualify the CS signal of an MC68881/MC68882 with AS (allowing 7 ns for a gate delay) and still meet the CS to DS setup time requirement (specification 8B of MC68881UM/AD, MC68881/MC68882 Floating-Point Coprocessor User's Manual).



*For the MC68EC020, A23-A0.
 This signal does not apply to the MC68EC020.
 NOTE: Timing measurements are referenced to and from a low voltage of 0.8 V and a high voltage of 2.0 V, unless otherwise noted. The voltage swing through this range should start outside and pass through the range such that the rise or fall will be linear between 0.8 V and 2.0 V.

FIGURE 10-3
MC68020UM

Figure 10-3. Read Cycle Timing Diagram

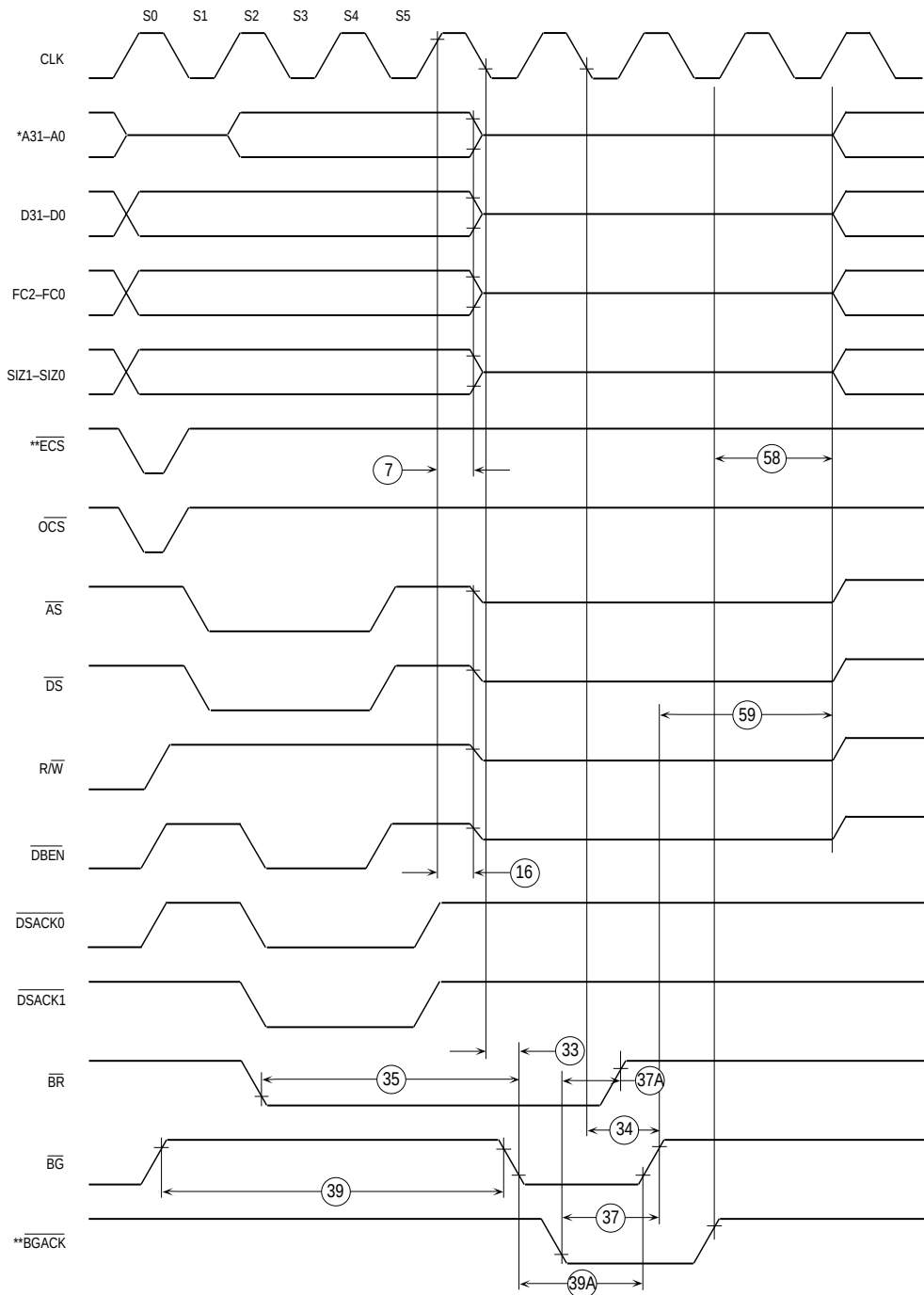


□ For the MC68EC020, A23-A0. □
 □ This signal does not apply to the MC68EC020.

NOTE: Timing measurements are referenced to and from a low voltage of 0.8 V □
 and a high voltage of 2.0 V, unless otherwise noted. The voltage swing □
 through this range should start outside and pass through the range such □
 that the rise or fall will be linear between 0.8 V and 2.0 V.
 □

FIGURE 10-4
 MC68020UM

Figure 10-4. Write Cycle Timing Diagram



*For the MC68EC020, A23-A0.

□ This signal does not apply to the MC68EC020.

NOTE: Timing measurements are referenced to and from a low voltage of 0.8 V and a high voltage of 2.0 V, unless otherwise noted. The voltage swing through this range should start outside and pass through the range such that the rise or fall will be linear between 0.8 V and 2.0 V.

FIGURE 10-5
MC68020UM

Figure 10-5. Bus Arbitration Timing Diagram

